

Lab Report: 1

Name: Saarthak Sabharwal

Roll number: 2023102055

Group number: 6

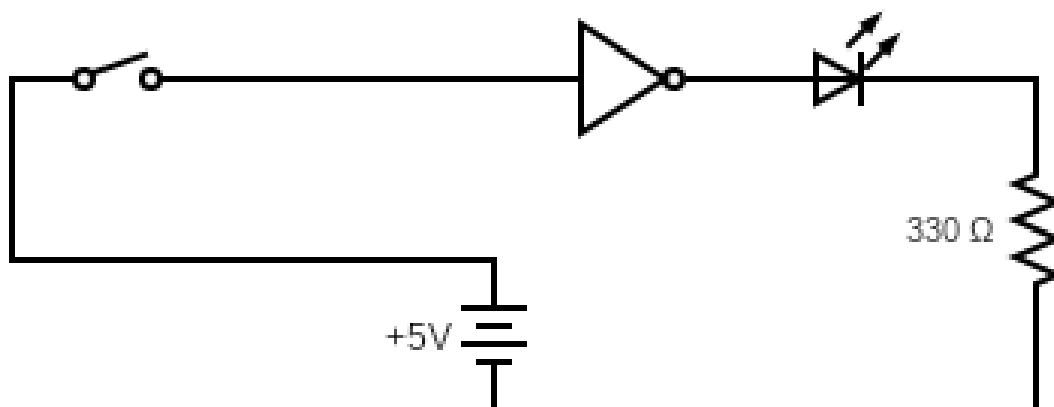
Experiment-1

Objective: Familiarization with Digital Test Kit and Binary Logic Levels

Electronic components used:

- Digital Test Kit (w/ breadboard)
- 74HC04 IC
- Wires

The reference circuit:



Procedure:

- Put the clock control switch in the FAST position and turn on the kit.
Note the glowing of the CLK-R LED. Now, put the clock control in SLOW position and notice the periodic blinking of the LED. Also, try working with the manual clock push button and notice the CLK-M LED.
- Verify the HIGH position of each input switch (whether toggle is up or down) by connecting them one by one to output LEDs and testing the toggle position for which the red LEDs glow.
- Connect the IC to the breadboard and power it by connecting pin 7 to ground and pin 14 to VCC.
- Make connections as shown in the reference diagram, with pin 1 as input to and pin 2 as output from the NOT gate. The input comes from one of the toggle switch outputs, IP1-IP12. The output goes to one of the display inputs, DP1-DP8.
- Note the input (from the switch) and the corresponding output (in the red/green LEDs).

Conclusion:

- The working of Digital Test Kit is now understood.
- The HIGH of the toggle switches are in the 'down' position.
- The NOT gate is realized with the help of an IC, namely, 7404.

Link for the Tinkercad simulation:

<https://www.tinkercad.com/things/gGwF5VMsVa4-copy-of-exp1/editel?sharecode=h2qFgL0FZgcQuzWZ-mZ0NVQDa0VEAIH0NB7UAKyesMM>

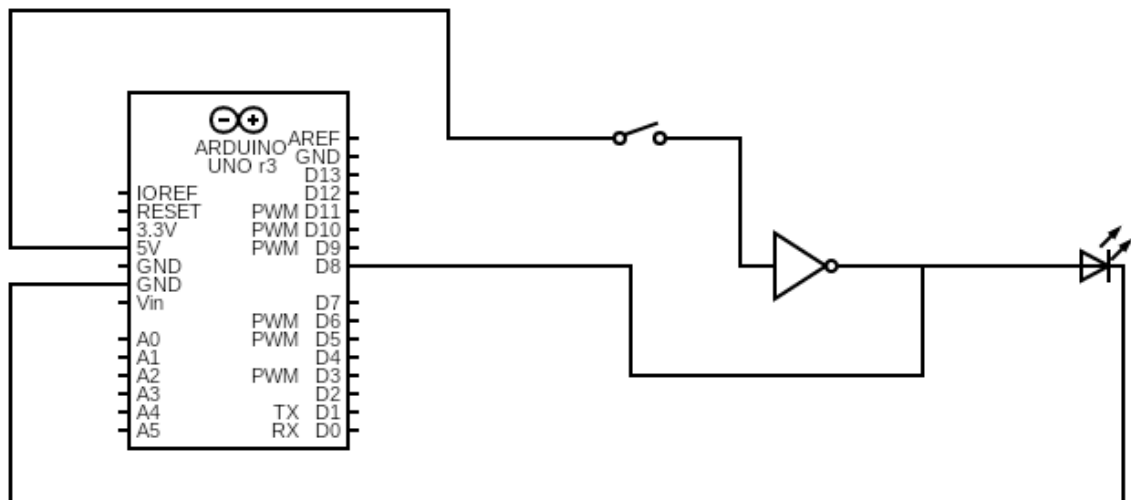
Experiment-2

Objective: Familiarization with Arduino and ATMEGA328P programming

Electronic components used:

- Digital Test Kit (w/ breadboard)
- Arduino Uno
- 74HC04 IC
- USB cable A to B
- Wires

The reference circuit:



Arduino Code:

```
1 //Auth: @Saarthak_2023102055
2 //reads digital input at pin 8 and prints it to serial monitor
3 //delay between consecutive toggles should at least be 500ms
4
5 void setup()
6 {
7     pinMode(8, INPUT);
8     Serial.begin(9600);
9 }
10
11 void loop()
12 {
13     if(digitalRead(8)) Serial.println("Hello World");
14     delay(500);
15 }
```

Procedure:

- Connect the IC to the breadboard, power it and make connections as shown in the reference diagram.
- Now, connect the Arduino to the circuit as well, as shown in the diagram.
- Make sure to power the circuit with the Arduino so as to ensure common ground voltage reference. This is to ensure that the logic levels are consistent throughout.
- Plug the USB into the board and PC. In Arduino IDE, select the board as “Arduino UNO”, select the relevant port, and upload the program shown above.
- Observe the output on the serial monitor as the input is flipped.

Conclusion: An understanding of basic Arduino has been acquired. The serial monitor outputs a ‘Hello World’ every 0.5s if the NOT gate output is HIGH, else nothing is printed.

Link for the Tinkercad simulation:

<https://www.tinkercad.com/things/7aVGSHR6Mq8-exp1/editel?sharecode=MFgy8LCniMufWUQNcpTJ9IqPQgddkKoVEXZf-WfZed8>